

Program to ADD 2 8-BIT data:

Memory address	Label	Instruction	OP code	Comments
8000		MVI C, 00	0E, 00	clear C reg to count carry
8002		LDA 4200H	3A, 00, 42	get 1 st data in A and store in B
8005		MOV B, A	47	
8006		LDA 4201H	3A, 01, 42	Get 2 nd data in A
8009		ADDB	80	Get Sum in A
800A		JNC B	02, 0E, 90	If CF=0 then skip next step
800D		INR C	0C	Increment C to count the carry
800E	SKIP	STA 4202H	82, 02, 42	Store the sum in memory
8011		MOV A, C	79	Move carry to acc and store in memory
8012		STA 4203H	82, 03, 42	
8015		HLT	76	stop

data:

Input	
Address	Data
4200	02H
4201	03H

Output	
Address	Data
4202	05H (sum)
4203	00H (carry)

P.No: 1

10.2023

8-BIT ADDITION WITH CARRY USING DIRECT ADDRESSING

Aim:

To write an assembly language program to add two numbers of 8-bit data stored in memory locations 4200H and 4201H and store the result in 4202H and 4203H with carry using direct addressing.

Apparatus Required:

1. 8085 microprocessor kit - 1
2. Power cord - 1
3. Keyboard - 1
4. 8085 Simulator and a PC

Algorithm:

1. Clear C reg to count carry.
2. Load the first data from memory to accumulator and move it to B register.
3. Load the Second data from memory to accumulator.
4. Add the content of B - register to accumulator.
5. Increment C register to count the carry
6. If carry flag = 0 then Turn to step 7
7. Store the Sum in memory
8. Move the carry to accumulator and store in memory
9. Stop.

Result:

Thus, an assembly language program for addition of given two 8-bit number with carry was written, executed and verified the Result successfully using

program to ADD two 16-bit data!

Memory address	label	Instruction	Op code	Comments
4100		MVI B, 00	06, 00	clean B reg to count carry.
4102		LHLD 4200H	2 A	Get 1st data in HL pair and save in DE
4103		XCHG	00	
4104		LHLD 4202H	42, E8	Get 2nd data in HL register
4106		BRD D	2 A, 02	ADD HL with DE and get the sum in HL register pair.
4108		JNC Skip	42, 19	If CY=0 then skip next step
410A		INR B	02, DE	Increment C register to count the carry.
410C	Skip	SHLD 4204H	41	Store the sum in memory
410D		MOV A, B	04	move the carry to acc & store in memory
410E		STA 4206H	22, 06, 42	
4111		HLT	78	Stop

data:

Input	
Address	Data
4200	04H
4202	08H

Output	
Address	Data
4204	0CH (sum)
4206	00H (carry)

16-BIT ADDITION WITH CARRY

Aim:

To write an assembly language program to add two numbers of 16-bit data stored in memory locations 4200H, 4201H, 4202H, & 4203H and store the results in 4204H, 4205H & 4206H with carry.

Apparatus Required:

1. 8085 microprocessor kit — 1
2. Power cord — 1
3. Keyboard — 1
4. 8085 Simulator and a PC.

Algorithm:

1. Clear B reg to count carry
2. Load the first data from memory to HL pair and move it to DE pair.
3. Load the second data from memory to HL pair.
4. Add the content of BC pair to HL pair.
5. If carry flag = 0 then jump to step 6
6. Increment D register to count the carry.
7. Store the sum in memory
8. Move the carry to accumulator and store in memory
9. Stop

Result:

Thus, an assembly language program for addition of given two ¹⁶ bit number with carry was written, executed and verified the Result successfully using 8085 Kit and Simulator.

Program to Subtract two 8-bit data

Memory Address	label	Instruction	Opcode	Comments
4100		LDA 4201H	3A, 01, 42	Get 2nd data in A and Save in B
4103		MOV B, A	47	Get
4104		LDA 4200H	37, 00, 42	Get 1st data in A register
4107		SUB B	90	Subtract B from A register
410A		JNC SKIP	02, 06, 41	If CY=0 then skip next 2 steps
410B		INR C	0C	Increment C to count the carry
410C		CMA	27	Take 2's complement of diff
410D		INR A	3C	
410E	Skip	STA 4202H	32, 02, 42	Store the difference in memory
4111		MOV A, C	79	Move the Borrow to accumulator and store in memory
4112		STA 4203H	32, 03, 42	
4115		HLT	76	Stop

Data:

Input	
Address	Data
4200	02H
4201	03H

Output	
Address	Data
4202	01H (Sum)
4203	00H (Carry)

Exp. no: 3

16.10.2023

8-BIT SUBTRACTION WITH BORROW USING DIRECT ADDRESSING

Aim:

To write an assembly language program to subtract two numbers of 8-bit data stored in memory locations 4200H and 4201H and store the result in 4202H and 4203H with borrow using direct addressing.

Apparatus Required:-

1. 8085 microprocessor kit — 1
2. Power cord — 1
3. Key board — 1
4. 8085 Simulator and a PC

Algorithm:

1. Load the second data from memory to accumulator and move it to B register.
2. Load the first data from memory to accumulator.
3. Subtract the content of B - register from accumulator.
4. If carry flag = 0 then jump to step 5.
5. Increment C register to count the borrow.
6. Take Two's complement of the difference.
7. Store the difference in memory.
8. Move the borrow to accumulator and store in memory.
9. Stop.

Result:

Thus, an assembly language program for subtraction of given two 8-bit number with borrow was written, executed and verified the Result successfully using 8085 Kit.

Program to Multiply 2 numbers of 8-bit data:

Memory address	label	Instruction	Opcodes	Comments
5000		LDA 4200H	3A, 0, 14, 2	Get 1 st data in A
5003		MOV E/A	5F	Save 1 st data in E
5004		LDA 4201H	3A, 14, 10, 1	Get 2 nd data in A
5007		MOV B/A	4F	Save 2 nd data in B
5008		LXI H, 0000H	21, 00, 00, 00	Load HL pair
500B		MVI D, 00H	16	Clear E for accounting overflow
500C	NEXT:	DAD D	00	Add the content of DE to sum
500D		DCR B	19	Decrement data 2 for every addition
500E		JNZ NEXT	05, 12	Repeat Addition until count is zero
5010		SHLD 4203H	0B, 50, 22	Store the product in memory.
5013		HLT	02	Stop

data:

Address	Input data	Address	Output data
4200	02H	4202	0CH
4201	06H	4203	00H

8-BIT MULTIPLICATION OPERATIONS USING 8085 MICROPROCESSOR

Aim:

To write an assembly language program to multiply two numbers of 8-bit data stored in memory 4200H and 4201H and store the product in 4202H and 4203H.

Apparatus Required:

1. 8085 microprocessor kit - 1
2. Power Cords - 1
3. Keyboard - 1
4. 8085 Simulator and a PC

Algorithm:

- * Load the first data in ACC and move to E.
- * Load the Second data ACC and move to B (count)
- * Clear HL pair (Initial sum)
- * Clear D for overflow (carry)
- * Add the content of DE to HL
- * Decrement the count.
- * Check whether count have reached Zero
- * Check the zero flag. If $ZF = 0$, repeat addition else if $ZF = 1$, go to next step.
- * Store the content of HL in memory. (Least Significant 16bits of the product).
- * Stop.

Result:

Thus, an assembly language program to multiply two numbers of 8-bit data was written, executed and verified the Result successfully using 8085 Kit.

Program to divide two numbers of 8-bit data:

Memory address	label	Instruction	Op code	Comments
4000		LDA 4201H	00, 8A, 0B	Get the divisor in B register
4003		MOV B, A	8A	Get the dividend in A register
4004		LDA 4200H	0B, 8A, 0B	Clear A register for quotient
4007		MVI C, 04H	A2, 09	
4009	AGAIN:	CMP B	08	If divisor is less than divided go to store
400A		JC STORE	0B, 8B, 0B,	
400D		SUB B	8A	Subtract divisor from divided
400F		INC C	0B	Increment quotient by one if each subtraction
400F		JMP AGAIN	8A, 0B	
4012		STA 4203H	0B, 8A, 0B, 1A	store the remainder in memory
4015		MOV A, C	8A	
4016		STA 4202H	0B, 8A, 0B	Store the quotient in Memory
4019		HLT	8A	Stop

data:

Address	Input data
4200	08 H
4201	02 H

Address	Output data
4202	02 H
4203	00 H

EXP. No. 5 8-BIT DIVISION OPERATIONS USING 8085 MICROPROCESSOR

16.10.2023

Aim:

To write an ALP to perform division of two 8 bit numbers stored in memory location 4200H, 4201H and store the remainder in 4202H and the quotient in 4203H.

Apparatus Required:

1. 8085 microprocessor Kit - 1
2. Power cord - 1
3. Keyboard - 1
4. 8085 Simulator and a PC

Algorithm:

1. Load the divisor in accumulator and move it to B-register
2. Load the dividend in accumulator.
3. Clear C-register to account for quotient
4. Check whether divisor is less than dividend
5. If divisor is less than dividend, go to Step 9, otherwise go to next step.
6. Subtract the content of B-register (quotient)
7. Increment the content of C-register (quotient)
8. Go to Step 4
9. Store the content of accumulator (remainder) in memory
10. Move the content of C-register (quotient) to accumulator and store in memory.
11. Stop.

Result:

Thus, an ALP to divide two numbers of 8-bit data was written, executed and verified the Result successfully using 8085 kit.

Program:

Address	label	Instruction	Opcode	Comments
1000		MOV CX, 0000H	C7, C1, 00, 00	Initialize counter CX
1004		MOV AX, [1200H]	8B, 06, 00, 12	Get First data in Ax
1008		MOV BX, [1202H]	8B, 0E, 02, 12	Get 2 nd data in Bx
100C		ADD AX, BX	01, D8	Add contents of Ax, Bx
100E		JNC L1	73, FF	Check for carry
1010		INC CX	41	If carry exists increment cx
1011	L1	MOV [1206H], CX	89, 0E, 06, 12	Store the carry
1015		MOV [1204H], AX	89, 06, 04, 12	Store the sum
1019		HLT	F4	Stop.

Output:

Input		Output	
Address	data	Address	data
1200	A8 42	1204	70 F0
1202	24 B4	1206	00 F0

Exp no: 6
17.10.2023

16-BIT ADDITION USING INDIRECT ADDRESSING WITH CARRY IN 8086 MICROPROCESSOR

AIM:

To write and execute an assembly language program to add two 16-bit unsigned numbers with carry in 8086 Kit and Emulator 8086.

Apparatus Required:

1. 8086 microprocessor kit — 1
2. Power cord — 1
3. Keyboard — 1
4. Emulator 8086 and a PC

Algorithm:

1. Load the First data in AX-register
2. Load the First data in BX-register
3. Add the two data and get the sum in AX-register.
4. If $C=0$ then skip next step.
5. Increment CX Reg for carry.
6. Store the sum in memory locations.
7. Store the carry in memory locations.
7. Stop the program.

Result:

Thus, an assembly language program for addition with carry of given 16-bit numbers was written, executed and verified the Result successfully using 8086 Kit and Emulator 8086.

ogram:

Address	Label	Instruction	OP Code	Comments
1000		MOV CX, 0000H	17, C1, 00, 00	Initialize counter CX
1004		MOV AX, [1300H]	8B, 06, 00, 13	Get 1 st data in AX
1008		MOV BX, [1302H]	8B, 1E, 02, 13	Get 2 nd data in BX
100C		SUB AX, BX	29, D8	Subtract contents of AX, BX
100E		JNC SKIP	73, 03	Check borrow
1010		INC CX	43	If carry exists, Increment the CX
1011		NEG AX	F7, D8	Take 1's Complement
1013	SKIP	MOV [1306H], CX	89, 0E, 06, 13	Store borrow
1017		MOV [1308H], AX	89, 06, 08, 13	Store difference
101B		HLT	F4	Stop

Output:

Input		Output	
Address	Data	Address	Data
1300	42	1306	00 00
1302	42 C2	1308	5F 80

p.no : 7

10.10.2023

SUBTRACTION OF 16-BIT NUMBERS WITH BORROW.

Aim:

To write and execute an assembly language program to subtract 16-bit unsigned numbers with borrow in 8086 Kit and Emulator 8086

Apparatus Required:

1. 8086 microprocessor Kit - 1
2. Power cord - 1
3. Keyboard - 1
4. Emulator 8086 and PC

Algorithm:

1. Load the Second data from memory to accumulator and move it to B register.
2. Load the First data from memory to accumulator.
3. Subtract the content of B-register from accumulator.
4. If carry flag = 0 then jump to step 5 & 6
5. Increment C register to count the borrow.
6. Take two's Complement of the difference.
7. Store the difference in memory.
8. Move the borrow to accumulator and store in memory
9. Stop.

Result:

Thus, an assembly language program for subtraction with borrow of given 16-bit numbers was written, executed and verified the Result successfully using 8086 Kit and Emulator 8086

Program:

Address	Label	Instruction	OP codes	Comments
1000		MOV AX, [1200H]	8B, 06, 00, 12	load AX register with 1 st data
1004		MOV BX, [1202H]	8B, 1E, 02, 12	load BX register with 2 nd data
1008		MUL BX	F7, E3	mult. the contents AX with BX
100A		MOV [1204H], AX	99, 06, 04, 12	Store the lower word
100E		MOV [1206H], BX	99, 1E, 06, 12	Store the higher word
1012		HLT	F4	Stop

Output:

Input		Output	
Address	Data	Address	Data
1200	06 04	1204	35 12
1202	08 07	1206	00 18

P.no: 8

2.10.2023

MULTIPLICATION OF 16-BIT NUMBERS

Aim:

To write and execute an assembly language program to Multiply two 16-bit unsigned numbers in 8086 Kit and Emulator 8086.

Apparatus Required:

1. 8086 Microprocessor Kit - 1
2. Power card - 1
3. Keyboard - 1
4. Emulator 8086 and PC.

Algorithm:

1. Load the Multiplier from memory to accumulates.
2. Load the Multiplicand from memory to BX Reg.
3. Multiply AX and BX.
4. Store the Lower word in memory from AX.
5. Store the Higher word in memory from DX.
6. Stop.

Result:

Thus, an assembly language program for Multiplication of given 16-bit numbers was written, executed and Verified the Result successfully using 8086 Kit and Emulator 8086.

program:

Address	label	Instruction	OP code	Comments
1000		MOV DX, [1200H]	8B, 16, 00, 12	Initialize DX register
1004		MOV AX, [1202H]	8B, 06, 02, 12	load AX register
1008		MOV BX, [1204H]	8B, 1E, 04, 12	Load BX register
100C		DIV CX	F7, F1	divide AX by BX
100E		MOV [1206], AX	89, 06, 06, 12	Store the Quotient
1012		MOV [1208], BX	89, 1E, 08, 12	Store Remainder
1016		HLT	F4	Stop

Input		Output	
Address	data	Address	data
1200	06	1206	0003
1202	00	1208	0000
1204	0002		

P.no: 9

10.23

DIVISION OF 32 BIT BY 16 BIT NUMBER

Aim:

To write and execute an assembly language program to Divide two 16-Bit unsigned numbers in 8086 Kit and Emulator 8086.

Apparatus Required:

1. 8086 microprocessor Kit - 1
2. Power card - 1
3. Keyboard - 1
4. Emulator 8086 and PC

Algorithm:

1. Load the Divisor from memory to accumulator
2. Load the Divisor from memory to BX reg
3. Divide DX:AX by BX.
4. Store the Quotient in memory from AX
5. Store the Remainder in memory from DX.
6. Stop.

Result:

Thus, an assembly language program for Division of given 16-bit numbers was written, executed and Verified the Result Successfully using 8086 Kit.

Program:

Address	label	Instruction	OP code	Comments
1000		MOV AX, [1200]	A1, 06, 12	Load AX register with 1 st data
1004		AND AX, 0F0FH	25, 0F, 0F	AND the content of Ax
1007		MOV [1202], AX	A3, B2, 04	store the result
1009		HLT	F4	Stop

Input		Output	
Address	Data	Address	Output data
1200 H	4F 5C	1202 H	0F 0F

Exp. no.: 10

7.10.2023

LOGICAL OPERATION

Aim:

To write and execute an assembly language program for performing Masking, Setting, One's and Two's complement of given 16-bit numbers

Apparatus Required:

- ✧ 8086 microprocessor kit — 1
- ✧ Power cord — 1
- ✧ Keyboard — 1
- ✧ Emulator 8086 and PC

Masking of Bits:-

Algorithm:

- ✧ Load the data in AX-register
- ✧ Logically AND the content of AX with 0F0FH
- ✧ Store the result in memory location
- ✧ Stop the program.

Program:

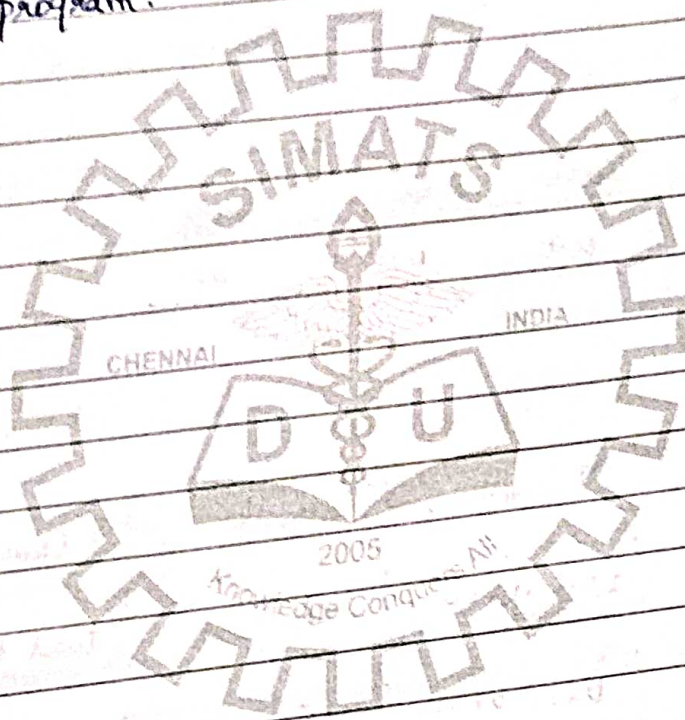
Address	label	Instructions	OPcode	Comments
1000		MOV AX, [1200]	A1, 00, 11	Load AX with 12 data
1004		OR AX, 0F0FH	00, 0F, 0F	OR the content of AX
1007		MOV [1202], AX	A3, B2, 04	Store No result
100B		HLL	Fu	Stop

Input		Output	
Address	data	Address	data
1200H	4C5D	1201H	4F5F

Setting of Bits:

Algorithm:

1. Load the data in AX - register
2. logically OR the content of AX with 0F 0F H
3. Store the result in memory location
4. Stop the program.



Result:

Thus, an assembly language program for performing logical Masking and Setting of bits were executed using K&T.

Program:

Address	label	Instruction	OPcode	Comments
4100		LDA 4200	3A,00,42	load the A register with data
4103		ANI 0FH	E6,0F	AND the content of A with 0F
4105		STA 4201	52,01,42	Store the result
4107		HLT	F4	Stop

Output:

Input		Output	
Address	data	Address	data
4200 H	4A	4201 H	0A

Program ii)

Address	label	Instruction	OPcode	Comment
4100		LDA 4200	3A,00,42	load the A register with data
4103		ORI, 0FH	E6,0F	OR the content of A with 0F
4105		STA 4201	52,01,42	Store the result
4107		HLT	F4	Stop

Output:

Input		Output	
Address	data	Address	data
4200 H	05	4201 H	0F

XP.No: 11

7.10.2023

LOGICAL OPERATION USING 8085 MICROPROCESSOR

Aim: To write and execute an ALP for performing Masking, setting, one's and Two's complement of given data of 8-bit numbers using 8085 Microprocessor.

Apparatus Required:

- * 8085 Microprocessor kit - 1
- * Power cord - 1
- * Keyboard - 1
- * 8085 Simulator and a PC

i) Masking of Bits:

Algorithm:

- * Load the Data in A register
- * Logically AND the content of A with OFH
- * Store the result in Memory location
- * Stop.

ii) Setting of Bits:

Algorithm:

- * Load the data in A-register
- * Logically ORI the content of A with OFH
- * Store the result in Memory location
- * Stop.

Result:

Thus an assembly language program for performing logical Masking and setting of bits were executed using 8085 kit

Program:

Address	label	Instruction	Op Code	Comment
4100		LDA 4200	7A, 00, 42	Load AX register with data
4103		CMA	2F	NOT the content of AX
4104		STA 4201	32, 01, 42	Store one's complement in memory
4107		INR A	3C	Increment the content of AX
4108		STA 4202	32, 02, 42	Store 2's complement in memory
410B		HLT	E4	Stop

Output:

Input		Output	
Address	data	Address	data
4200 H	AB	4201 H	54
		4202 H	55

iii) One's Complement and Two's Complement

Aim:

To write and execute the assembly language program for performing one's and Two's complement of given 8-bit numbers using 8085 Microprocessor.

Apparatus Required:

- * 8085 Microprocessor Kit - 1
- * Power cord — 1
- * Key board — 1
- * 8085 Simulator and a PC

Masking of Bits:

Algorithm:

- * Load the data in A-register
- * Logically NOT the content of A
- * Store the 1's complement in Memory location
- * Increment the content of A
- * Store the 2's complement in Memory
- * Stop.

Result:

Thus an assembly language program for performing One's & Two's complement of bits were executed using 8085 Kit.

Program:

Address	label	Instruction	Opcode	Comments
4100		MOV A, #data	74, 05	load data in acc
4102		MOV B, #0F	74, F0, 05	Load data
4107		ORL A, B	35, F0	OR the contents of acc with reg
4109		MOV DPTR, #4500H	90, 11, 00	Initialize DPTR with 4500
410A		MOVX @DPTR, A	. F0	Store the result
410A	stop	SJMP STOP	50, FF	Stop

Output

Input		Output	
Address	data	Address	data
4101	13	4500	CF

Program:

Address	label	Instruction	Opcode	Comments
4100		MOV A, #data	74, 05	load the data in acc
4102		MOV B, #0F	75, F0, 05	load the data in B reg
4105		ANL A, B	35, F0	AND the content of acc with B
4107		MOV DPTR, #4500	90, 11, 00	Initialize DPTR with 4500
4109		MOVX @DPTR, A	. F0	.
410A	STOP	SJMP STOP		stop

Input		Output	
Address	Data	Address	data
4101	4D	4500	0D

EXP. NO: 12

LOGICAL OPERATIONS USING 8051 MICROCONTROLLER

7.10.2023

Aim:

To write and execute an assembly language program for setting and Masking of given 8-bit numbers using 8051.

Apparatus Required:

- * 8051 Microcontroller Kit - 1
- * Powercard - 1
- * Keyboard - 1
- * Keil and PC

Setting of Bits:

Algorithm:

- * Load the data in A register
- * Load 0F to set the lower nibble in B
- * Perform OR operation with B
- * Store the Result in memory
- * Stop the program.

Masking of Bits:

Algorithm:

- * Load the data in A register
- * Load 0F to mask the higher in B
- * Perform AND operation with B
- * Store the Result in Memory
- * Stop the Program.

Result:

Thus an Assembly Language program for setting and Masking of 8-bit numbers using 8051 Microcontroller were performed and its output were verified.

Program:

Address	label	Instructions	OP code	Comments
4100		MOV A, #data	74, 05	load the data 1 in accumulator
4102		MOV B, #data	74, F0, 05	load the data 2 in reg B
4105		ADD C A, B	35, F0	Add the contents
4107		MOV OPTR, #4500H	90, 11, 00	Initialize OPTR with 4500H
410A		MOVX @OPTR, A	F0	Store the result
410B	stop	SJMP STOP	80, FF	stop

Output:

Input		Output	
Address	data	Address	data
4101	05	4500	09
4104	04		

ADDITION OPERATION USING 8051 MICROCONTROLLER

Aim:

To write and execute an assembly language program to Add of two 8-bit numbers using 8051 and Keil software.

Apparatus Required:

- * 8051 Micro controller Kit - 1
- * Powercord - 1
- * Keyboard - 1
- * Keil and PC

Algorithm:

- * Load the first data in A-register
- * Load the Second data in B-register
- * Add the two data with carry
- * Store the sum in Memory location
- * Stop the program.

Result:

Thus an assembly language program for addition of given two 8-bit numbers was written, executed and verified the Result successfully using 8051 Kit.

Program

Address	label	Instructions	Comment
4100		MOV A, #data	Load 1 st data in A
4102		MOV B, #data	load 2 nd data in B
4105		SUBB A, B	Subtract Contents of A and B
4107		MOV DPTR, #4500H	Initialize DPTR with 4500H
410A		MOVX @DPTR, A	Store the result
410B	STOP	STOP STOP	STOP

Output

Input		Output	
Address	data	Address	data
4101	03	4500	01
4104	07		

XP-NO: 114

10.10.2023

SUBTRACTION USING 8051 MICROCONTROLLER

Aim:

To write and execute an assembly language program to subtract two 8-bit numbers using 8051.

Apparatus Required:

- * 8051 Microcontroller kit - 1
- * Powersupply - 1
- * Keyboard - 1
- * Keil and PC

Algorithm:

- * Load the first data in A-register
- * Load the second data in B-register
- * Subtract the two data with borrow
- * Store the Sum in Memory location
- * Stop the program.

Result:

Thus an assembly language program for subtraction of given two 8-bit number was written executed and verified the Result successfully using 8051 Kit.

Program:

Address	label	Instruction	Comments
4100		MOV A, #data	load 1 st data in A
4102		MOV B, #data	load 2 nd data in B
4105		MUL A, B	Multiply contents A & B
4106		MOV DPTR, #4500H	Initialize DPTR with 4500H
4109		MOV X@DPTR, A	Store the result
410A		INC DPTR	Increment Data pointer
410B		MOV A, B	Copy contents of B to A
410D		MOV X@DPTR, A	Store MSB in 4501
410E	STOP	STMP STOP	Stop

Output:-

Input		Output	
Address	Output	Address	Output
4101	03	4500	06
410A	02	4501	00

Exp. No.: 15

18.10.2023

MULTIPLICATION USING 8051 MICROCONTROLLER

Aim:

To write and execute an assembly language program to multiply two 8-bit numbers using 8051.

Apparatus Required:

- * 8051 Microcontroller - 1
- * Powercard - 1
- * Keyboard - 1
- * Keil and PC

Algorithm:

- * Get the multiplier in the accumulator
- * Get the multiplicand in the B
- * Multiply A with B
- * Store the result in Memory
- * Stop the program.

Result:

Thus an assembly language program for multiplication of given two 8-bit number was written, executed and verified the result successfully using 8051 kit.

Program:

Address	Label	Instructions	Comments
4100		MOV A, #data	Load 1 st data in A
4102		MOV B, #data	Load 2 nd data in B
4105		DIV AB	Divide contents of A and B
4106		MOV DPTR, #4500H	Initialize DPTR with 4500H
4109		MOVX @DPTR, A	Store quotient in 4500H
410A		INC DPTR	Increment DPTR
410B		MOV A, B	Copy contents of B to A
410D		MOVX @DPTR, A	Store the remainder in 4501H
410E	STOP	SJMP STOP	Stop

Output:

Input		Output	
Address	Data	Address	Data
4101	08	4500	04
4104	02	4501	00

DIVISION USING 8051 MICRO CONTROLLER

EXP. No. 16
19/10/2023

Aim:

To write and execute an assembly language program to divide two 8-bit numbers using 8051.

Apparatus Required:

- * 8051 Microcontroller Kit - 1
- * Powercord - 1
- * Keyboard - 1
- * Keil and PC

Algorithm:

- * Get the dividend in the accumulator
- * Get the divisor in the B-register
- * Divide A by B
- * Store the Quotient and Remainder in memory
- * Stop the program.

Result:

Thus an assembly language program for division of given two 8-bit numbers was written, executed and verified the Result successfully using 8051 kit.

Program:

Address	OpCode	Label	Instruction	Comments
1000	C7, C6, 00, 12		MOV SI, 1150 H	Initialize the source address
1004	C7, C7, 00, 13		MOV DI, 1250 H	Initialize the destination address
1006	C7, C12, FC, 06, 00		MOV CX, 0006 H	Initialize count value to count register
1009	FC	REPEAT	CLD	Clear the direction Flag
100A	AC,		MOV SB	Move the String Byte
100B	F2, F3		LOOP REPEAT	Unconditional loop to address loop to address
100D	F4		HLT	Stop

Input		Output	
Address	Data	Address	Data
1150	52 H	1256	52 H
1151	53 H	1251	53 H
1152	54 H	1252	54 H
1153	55 H	1253	55 H
1154	56 H	1254	56 H

MOVE A DATA BLOCK WITHOUT OVERLAP

Exp No: 17
19.10.23

Aim:

To write and execute an assembly language program for transferring data from one block to another block without overlapping using 8086 Kit and Emulator 8086.

Apparatus Required:

- * 8086 Microprocessor Kit - 1
- * Power Card - 1
- * Keyboard - 1
- * Emulator 8086 and PC

Algorithm:

- * Initialize counter
- * Initialize source block pointer
- * Initialize destination block pointer
- * Get the byte from source block
- * Store the byte in destination block
- * Increment source, destination pointers and decrement counter
- * Repeat Steps 4, 5 and 6 until counter equal to zero
- * Stop.

Result:

Thus, an assembly language program for transferring data from one block to another block without overlapping was executed using 8086 Kit.

Program.

Address	Label	Instruction	OPCode	Comment
4100	START	LXI H, 4200	21	Initialize HL with 4200H
4103		MVI C, 04	06	Copy the value 04 to C
4105	NEXT	MOV A, M	7E	Copy the content M to A
4106		OUT C	D3	Content of A moved to O/P
4108		LXI D, 1010	D3	Copy the data 1010 to DE
410B	LOOP	DCX D	11	Decrement DE - reg
410C		MOV A, E	1B	Check DE - 0000
410D		DRA D	7B	Jump on no zero to loop
410E		JNC LOOP	B2	Increment HL reg pair
4111		INX H	C2	Decrement the count
4112		DCR C	23	
4113		JNC NEXT	05	Jump to next if Z flag is 0
4115		JMP START	C2	Jump to label start
4118		HLT	C3, 06	Stop
4200		09 05 06 0A	09, 05	Clockwise direction
4200		0A 06 05 09	F4	Counter clockwise direction.

Output:

Input		Output
address	data	Motor rotates on
4200	09 05 06 0A	Clockwise direction
4200	0A 06 05 09	Counter clockwise direction

STEPPER MOTOR INTERFACING

EXP. No: 17

9-10-23

Aim:

To write and execute an assembly language program to run a stepper motor at different speed, and to control its direction using 8085 Microprocessor.

Apparatus Required:

- * 8085 Microprocessor Kit - 1
- * Stepper Motor - 1
- * Stepper Motor Interface board - 1
- * Power card - 1
- * Keyboard - 1

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Result:

Thus an assembly language program to run the stepper motor in both forward and reverse direction with delay was executed and its output was verified.

Address	OPcodes	Label	Instructions	Comments
1000	C7, C6, 00, 12	START	MOV SI, 1200H	Initialize array
1004	C7, C6, 00, 0F		MOV CX, 000F	Initialize array :2c
1008	C6, C0, 10		MOV AL, 10	Store the control word
100B	E6, C2		OUT C2, AL	Store through output port
100D	C6, C0, 1C		MOV AL, 1C	Store control word to delay
1010	E6, C2		OUT C2, AL	Send through output port
1012	C6, C0, 90		MOV AL, 90	sent through out port
1015	E6, C2		OUT C2, AL	Send through first data
1017	9A, C0, 4		MOV AL, 4	Store in move to AL
1019	E6, C0	NEXT	OUT C0, AL	Sent through output port
101B	E3, E4, 04		CALL DELAY	Call delay
101E	16		INC SI	Increment the source
101F	E2, 1F, 6		LOOP NEXT	Index
1027	E9, D0, 1F		JUMP START	Jump to start
1500	C7, C2, FF, A0	DELAY	MOV DX, 0A0FF	MOV 0A0FF to DX
1564	C6, 4	LOOP	DEC DX	Decrement DX
1505	75, FD		JNC LOOP	Jump to loop
1507	C3		HLT	Stop

Input

1200	98H	68H	7CH	E8H
1204	1FH	1CH	29H	FFH

KEYBOARD AND DISPLAY

Exp No. 20
19.10.23

Aim:

To write and Execute an assembly language program to roll message "HELP US" in the display

Apparatus Required:

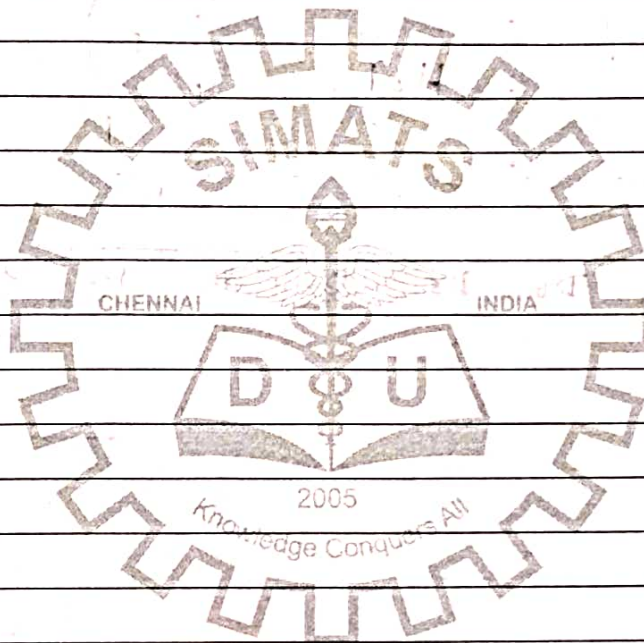
- * 8086 Microprocessor kit - 1
- * 8279 interface board - 1
- * Power cord - 1
- * Keyboard - 1

Algorithm:

1. Initialize the counter
2. Set 8279 for 8-bit digit character display
3. Set 8279 for clearing the display
4. Write the commands to display
5. Load the character into accumulator and display it
6. Introduce the delay
7. Repeat from Step 1

Memory location	Message	T-segment LED Format								Hexa code
		D	C	B	A	DP	G	F	E	
1200H	H	1	0	0	1	1	0	0	0	98H
1201H	F	0	1	1	0	1	0	0	0	68H
1202H	L	0	1	1	1	1	1	0	0	7CH
1203H	P	1	1	0	0	1	0	0	0	68H
1204H	-	1	1	1	1	1	1	1	1	FFH
1205H	V	0	0	0	0	1	1	0	0	1CH
1206H	S	0	0	1	0	1	0	0	1	29H
1207H	-	1	1	1	1	1	1	1	1	FFH

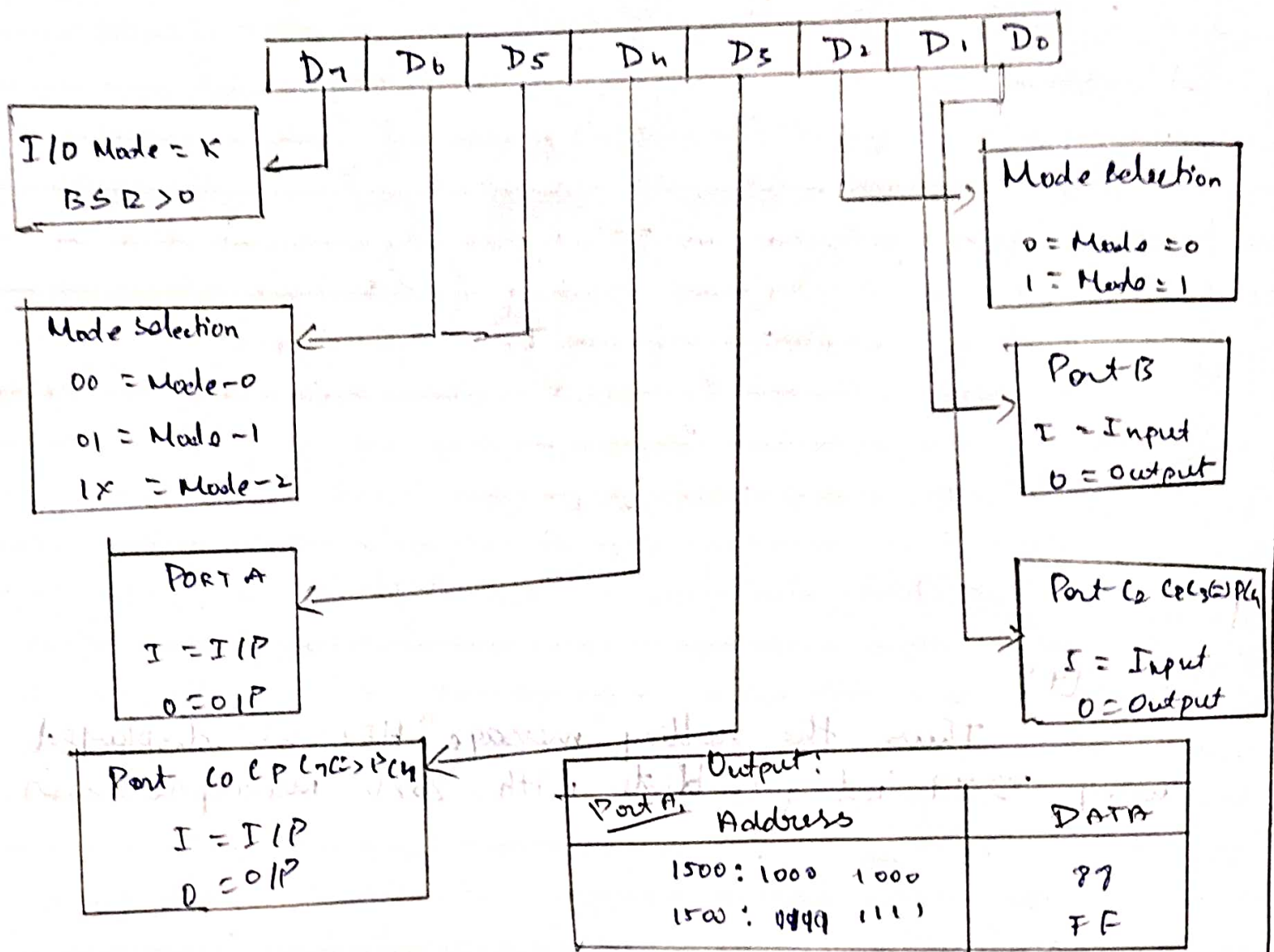
The first two rows are identical.
 The third row is identical to the first row.
 The fourth row is identical to the first row.
 The fifth row is identical to the first row.
 The sixth row is identical to the first row.
 The seventh row is identical to the first row.
 The eighth row is identical to the first row.



Result:

Thus the rolling message "HELP US" displayed using 8279 interface with 8086 microprocessor.

Memory Address	LABEL	Assembler	OPCode	Comments
1000		MOV AL, 90H	B0, 90	Load the AL with control word
1002		OUT CB, AL	E6, AL	Send the control word to control
1004		IN AL, C0	F4, C0	Read port A
1006		MOV [1500], AL	87, 04	Store the result on memory
1009		HLT	F4	Stop



EXP. no. 2

19.10.2022

INTERFACE SWITCHES WITH 8086 THROUGH 8255

Aim:

To write and execute an ALP to interface switches with 8086 through 8255 PPI

Apparatus required:

- * 8086 Microprocessor Kit - 1
- * 8255 Interface Board - 1
- * Power cord - 1
- * Keyboard - 1

Algorithm:

1. Configure the 8255 port A as input with the control reg value as "90H"
2. Read the port A switch status through C.
3. Store the output in 1250
4. Stop

Result:

Thus, an ALP for Interfacing of switches with 8086 through 8255 PI was written executed and verified the result successfully.